BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8487

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. During nitration of benzene, the active nitrating agent is:

(A) NO_2 (B) HNO_3

(C) NO (D) NO_2^+

Q2. For which Mechanism, the first step involved is the same?

(A) E1 and E2 (B) E2 and S_N2

(C) S_N1 and E2 (D) E1 and S_N1

Q3. Which enzyme is not involved in fermentation of starch?

(A) Diastase (B) Zymase

(C) Urease (D) Invertase

Q4. Cannizzaro's reaction is not given by:

(A) Acetaldehyde (B) Formaldehyde

(C) Benzaldehyde (D) Trimethyl acetaldehyde

Q5. Acetic acid was first isolated from:

(A) Milk
(B) Vinegar
(C) Butter
(D) Red ant

Q6. Which is an addition polymer?

(A) Nylon 6-6
(B) Terylene
(C) Polystyrene
(D) Epoxy resin

Q7. Phosphorus helps the growth of:

(A) Root
(B) Leave
(C) Stem
(D) Seed

Q8. The pH range of the acid rain is:

(A) 7 – 6.5
(B) 6.5 – 6
(C) 6 – 5.6
(D) Less than 5

Q9. Half of the atmospheric mass is concentrated in the lower:

(A) 5.6 km
(B) 10.6 km
(C) 15.6 km
(D) 20.6 km

Q10. Among alkali metal ions, minimum hydration energy is shown by:

(A) Li^+
(B) Na^+
(C) Rb^+
(D) K^+

Q11. The oxide of beryllium is:

(A) Acidic
(B) Basic
(C) Neutral
(D) Amphoteric

Q12. Which element is not present abundantly in earth's crust?

(A) Si
(B) Al
(C) Na
(D) O

Q13. Which catalyst is used to contact process?

(A) V_2O_5
(B) Fe_2O_3
(C) SO_3
(D) Ag_2O

Q14. Chlorine heptaoxide (Cl_2O_7) reacts with water to form:

(A) Hypochlorous acid
(B) Chloric acid
(C) Perchloric acid
(D) Chlorine and oxygen

Q15. Coordination number of Pt in [Pt Cl(NO_2)(NH_3)$_4$] is:

(A) 02 (B) 04

(C) 01 (D) 06

Q16. Which set of hybrid orbitals has planar triangular shape?

(A) sp^2 (B) sp^3

(C) sp (D) dsp^2

Q17. Vinyl acetylene combines with HCl to form:

(A) Poly acetylene (B) Chloroprene

(C) Benzene (D) Divinyl acetylene

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8487

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Why the size of cation is smaller than that of parent atom?

(ii) ZnO oxide is amphoteric oxide. Justify with two reactions.

(iii) Write any two uses of gypsum in industry.

(iv) Write two major problems during manufacturing of NaOH in diaphragm cell.

(v) Write four uses of borax.

(vi) Why are liquid silicones preferred over ordinary organic lubricants?

(vii) Write any four uses of H_2SO_4.

(viii) Write four similarities of oxygen and sulphur.

(ix) Why does damaged tin plated iron get rusted quickly?

(x) What is sacrificial corrosion?

(xi) What are fertilizers? Why they are needed?

(xii) Write any four essential qualities of good fertilizer.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Why iodine has metallic luster?

(ii) Give any four uses of bleaching powder.

(iii) Define functional group and give two examples.

(iv) What is metamerism? Give example.

(v) What is Clemmensen reduction? Give one example.

(vi) How is mustard gas prepared? Give its use.

(vii) How does ethyne react with water in the presence of $HgSO_4/H_2SO_4$

(viii) Write a method for preparation of ethyl magnesium bromide in laboratory.

(ix) What are β-eliminations reactions?

(x) Differentiate between conjugated and derived proteins.

(xi) Write structure of cholesterol.

(xii) Give two difference between DNA and RNA.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What happens when Cl_2 is passed through benzene in sunlight?

(ii) Convert benzene into glyoxal.

(iii) Give IUPAC name: (a) $CH_3 – CH(OH)COOH$ (b) $CH_2(OH) – CH_2(OH)$

(iv) Define fermentation. Give its necessary conditions.

(v) Convert ethanal into ethanol.

(vi) What are Zwitter ions? Give example.

(vii) What is glacial acetic acid? Why is it called so?

(viii) How acid rain affects our environment? Briefly discuss.

(ix) What are leachates? Briefly explain.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Discuss the position of hydrogen on top of group IA. (four similarities and four differences)

Q5(b). (Marks 04) Write applications of aluminium. (any eight)

Q6(a). (Marks 04) Write the formulas of these minerals: (i) Dolomite (ii) Asbestos (iii) Epsom salt (iv) Sylvite

Q6(b). (Marks 04) Define corrosion. Explain electrochemical theory of corrosion.

Q7(a). (Marks 04) What is orbital hybridization? Explain structure of ethane on the basis of sp^3-hybridization?

Q7(b). (Marks 04) Explain mechanism of S_N1 reactions with a suitable example.

Q8(a). (Marks 04) Describe the Kolbe's electrolytic method for the preparation of alkenes along with mechanism.

Q8(b). (Marks 04) Explain Cannizzaro's reaction with the help of mechanism of formaldehyde.

Q9(a). (Marks 04) What are Friedel craft alkylation reaction? Give its mechanism.

Q9(b). (Marks 04) Give the reaction of phenol with: (i) HNO_3 (ii) H_2SO_4 (iii) CH_3COCl (iv) Br_2 water

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8482

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Keeping in view the size of atoms, which order is the correct?

(A) Mg > Sr
(B) Ba > Mg
(C) Lu > Ce
(D) Cl > I

Q2. Which does not belong to alkaline earth metals?

(A) Be
(B) Ra
(C) Ba
(D) Rn

Q3. Which metal is used in the thermite process because of its activity?

(A) Iron
(B) Copper
(C) Aluminium
(D) Zinc

Q4. Laughing gas is chemically:

(A) NO
(B) N_2O
(C) NO_2
(D) N_2O_4

Q5. The anhydride of $HClO_4$ is:

(A) ClO_3
(B) ClO_2
(C) Cl_2O_5
(D) Cl_2O_7

Q6. Coordination number of Pt in [Pt Cl(NO_2)(NH_3)$_4$] is:

(A) 2
(B) 4
(C) 1
(D) 6

Q7. A double bond consists of:

(A) Two sigma bonds
(B) One sigma and one pi bond
(C) One sigma and two pi bonds
(D) Two pi bonds

Q8. Vinyl acetylene combines with HCl to form:

(A) Poly acetylene
(B) Benzene
(C) Chloroprene
(D) Divinyl acetylene

Q9. Benzene cannot undergo:

(A) Substitution reaction
(B) Addition reaction
(C) Oxidation reaction
(D) Elimination reaction

Q10. S_N2 reactions can be best carried out with:

(A) Primary alkyl halides
(B) Secondary alkyl halides
(C) Tertiary alkyl halides
(D) All these

Q11. Which enzyme is not involved in fermentation of starch?

(A) Diastase
(B) Zymase
(C) Urease
(D) Invertase

Q12. Cannizzaro's reaction is not given by:

(A) Formaldehyde
(B) Acetaldehyde
(C) Benzaldehyde
(D) Trimethyl acetaldehyde

Q13. Which reagent is used to reduce a carboxylic group to an alcohol?

(A) H_2/Ni
(B) H_2/Pt
(C) $NaBH_4$
(D) $LiAlH_4$

Q14. A polymeric substance that is formed in the liquid state and then hardened to a rigid solid is called a:

(A) Fiber
(B) Plastic
(C) Varnish
(D) Polyamide resin

Q15. Phosphorous helps the growth of:

(A) Root
(B) Leave
(C) Stem
(D) Seed

Q16. Fungicides are the pesticides which:

(A) Control the growth of fungus
(B) Kill insects
(C) Kill plants
(D) Kill herbs

Q17. In purification of potable water the coagulant used is:

(A) Nickle sulphate
(B) Copper sulphate
(C) Barium sulphate
(D) Alum

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8482

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Why metallic character increases from top to bottom in a group of metals?

(ii) What are polymeric halides? Give example.

(iii) How is lime mortar prepared?

(iv) Why is the aqueous solution of Na_2CO_3 alkaline in nature?

(v) How will you convert boric acid into borax and vice versa?

(vi) Why CO_2 is non-polar in nature?

(vii) What is meant by Fuming nitric acid?

(viii) NO_2 is a strong oxidizing agent. Prove the truth of this statement giving examples.

(ix) Why does damaged tin plated iron get rusted quickly?

(x) What are chelates? Give example.

(xi) Describe prilling of urea.

(xii) What do you mean by setting of cement?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) How ClO_2 is prepared? Give its reaction.

(ii) What are freons and teflons? Give their importance.

(iii) Define homologous series. Also give two examples.

(iv) Why do ethers and ketones show metamerism? Justify.

(v) Write structural formula of (a) vinyl bromide (b) 3-n-propyl-1, 4-pentadiene.

(vi) How will you prepare propene from isopropyl chloride?

(vii) Identify "A and B": $CH_3CH_2CH_2OH$ --PCl_5--> A --Na/Ether--> B

(viii) Convert methane into methanol.

(ix) Discuss the reactivity of alkyl halides.

(x) What is saponification number? Give saponification number of tripalmitate.

(xi) Explain the classes of enzymes with one example in each (a) isomerase (b) lyases.

(xii) Discuss the specificity of enzymes.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) How is m-chloronitrobenzene prepared from benzene?

(ii) Define resonance. Give one example.

(iii) How ethanol reacts with Conc·H_2SO_4 at different temperatures?

(iv) Write note on Lucas test.

(v) How will you distinguish between butanone and 3-pentanone?

(vi) Write four uses of acetic acid.

(vii) Define essential and non-essential amino acids.

(viii) What are leachates?

(ix) Define oxidizing and reducing smog.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) What is ionization encrgy? Give an example. How does it vary in group and periods?

Q5(b). (Marks 04) Write eight uses of borax.

Q6(a). (Marks 04) Compare the chemical behaviour of lithium with magnesium. (any four points)

Q6(b). (Marks 04) Describe the following properties of transition metals (i) Alloy formation (ii) Paramagnetism.

Q7(a). (Marks 04) Define sp-hybridization. Explain the structure of ethyne on the basis of sp-hybridization.

Q7(b). (Marks 04) Explain nucleophilic substitution bimolecular reaction. (S_N2)

Q8(a). (Marks 04) Explain the acidic character of alkynes with two examples.

Q8(b). (Marks 04) What is Cannizzaro's reaction? Explain with mechanism.

Q9(a). (Marks 04) Explain the terms with reference to alcohols: (i) Dehydration (ii) Oxidation

Q9(b). (Marks 04) Explain the rules for nomenclature of monocyclic aromatic hydrocarbons and their derivatives. (any four)